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Evaluating sound attenuation of single trees using 3D information

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Abstract

Urban tree belts reduce noise pollution, but limited research has focused on the mitigation potential of single trees. Identifying individual tree characteristics that influence noise propagation can assist in selecting trees to improve urban soundscapes at multiple scales. This study introduces a methodology to evaluate and predict the sound attenuation of single trees using 3D tree morphology data and sound observations. We extracted structural characteristics for 26 trees on Nanjing University's Xianlin campus from handheld terrestrial LiDAR. Second, the sound attenuation of each sample tree was quantified systematically using a sound source and a receiver. The sound level meter was placed in front of and behind each sample tree to record the received sound levels. The sound source was positioned 1.5m above ground to emit white noise, ensuring the front receiver recording sound levels of 55, 60, and 68 dBA. We established a support vector regression (SVR) with a linear (LN) kernel to predict the sound attenuation of single trees based on their 3D characteristics. Single trees yielded an insertion loss of 2-3 dBA, effectively eliminating sound above 500 Hz and increasing with the frequency. It is also interesting to note that the insertion loss increases with increasing source sound levels. Regression analysis revealed that an increase in crown leaf area index ($\beta = 0.332$, $p < 0.01$) and average leaf inclination ($\beta = 0.168$, $p < 0.01$) reduced sound significantly, indicating

the tree canopy's predominant role in impeding sound propagation. The SVR-LN model, established using standardized parameters with statistical significance, exhibited strong predictive sound attenuation performance using tree characteristics (R^2=0.74, RMSE=0.38, and MSE=0.15). This study addresses a research gap in the acoustic effects of single trees and provides a framework for accurately evaluating and predicting sound attenuation based on 3D characteristics. The findings can assist urban planners and policymakers in strategically planting trees to foster healthier and quieter living spaces for residents.

Keywords: sound attenuation, single trees, LiDAR, 3D tree characteristics, support vector machine prediction

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1 Introduction

In recent decades, noise pollution, as one of the primary environmental threats to public health, has increased in magnitude and distribution. According to the European Environment Agency (EEA, 2020), an estimated 113 million people were affected by long-term daily noise levels of at least 55 dB, while 79 million people were exposed to night-time noise levels of at least 50 dB. Accumulating evidence has confirmed prolonged noise exposure can lead to various adverse health outcomes, such as high blood pressure, cardiovascular disease, and suppressed immune function (Manuella Lech Cantuaria, 2023). Moreover, noise pollution can cause cognitive impairment, induce stress responses, and further alter the behavior of animals (Nelson-Olivieri et al., 2024). For example, animals living in areas exposed to anthropogenic noise suffer from reduced reproductive success and increased mortality (Lehnardt et al., 2024).

Multiple solutions are required to tackle this issue to achieve WHO's recommended standards, which suggest maximum noise levels of 53 dB during the day and evening and 45 dB at night (WHO, 2018). Additionally, one of the zero pollution targets for 2030 is to achieve a 30% reduction in the proportion of people chronically disturbed by transport noise compared to levels observed in 2017 (EC, 2021). Therefore, strategies for noise reduction have become increasingly and continuously important. Many technological approaches to noise attenuation, such as synthetic noise barriers and sound-absorbing materials, have been developed but can adversely impact urban ecosystems (Mitrus and Zbyryt, 2018). In recent years, urban green infrastructure (UGI) has emerged as an innovative and sustainable solution to alleviate noise. The acoustic effects of UGI can be divided into ground interference, sound absorption, and sound redirection (Van Renterghem et al., 2015). Specifically, porous surfaces beneath plants absorb and scatter incident sound and destructively interfere with direct and ground-reflected sound (Attenborough et al., 2016); vegetation absorbs noise by converting sound energy to heat energy via mechanical vibrations of plant elements and air particles (Romanova et al.,

2019); and vegetation redirects sound propagation through reflection, scattering, and diffraction (Yang et al., 2013a). In addition, UGI-based noise abatement solutions that introduce natural sounds and pleasant landscapes can eliminate noise annoyance (Kong et al., 2007) and improve public health (Buxton et al., 2021). Therefore, UGI is a viable noise abatement strategy for urban soundscape management.

An essential requirement for using UGI for noise reduction is identifying the vegetation parameters influencing acoustic effects. The normalized difference vegetation index (NDVI) is an indicator widely applied to monitoring vegetation coverage at regional scales and has been used to estimate noise levels in urban areas. Liu et al. (2020) demonstrated that noise levels decreased with increased NDVI. However, NDVI is a two-dimensional indicator commonly used to assess the qualitative relationship between noise levels and green spaces rather than a morphological parameter required to evaluate the impact of the three-dimensional (3D) vegetation structure on noise levels at small scales. Several studies focused on morphological parameters that determine the acoustic performance of UGI (Fang and Ling, 2003; Horoshenkov et al., 2013; Zhang et al., 2015; Zhao et al., 2021). A soil experiment with acoustic impedance tubes indicated that key parameters characterizing the acoustic effects of soils included porosity, pore size distribution, flow resistivity, and moisture content (Horoshenkov and Mohamed, 2006; Van Renterghem et al., 2021). It is worth noting that the ground effect is negligible when the sound source and receiver are close (Attenborough et al., 2014). Horoshenkov et al. (2013) used an equivalent fluid model to simulate sound propagation and revealed that the leaf inclination, total area and weight of leaves, and leaf area density of trees influence the thermoviscous absorption of sound. Fang and Ling (2003) investigated the noise reduction capacity of 35 tree belts and found that visibility, width, height, and length of tree belts are important noise reduction factors. However, traditional methods to sample vegetation morphology, such as image processing (Zhu and Chen, 2023), usually cannot capture detailed 3D information, leading to low accuracy in characterizing vegetation structures.

Light Detection and Ranging (LiDAR) has emerged as a novel and powerful technique to accurately extract 3D vegetation characteristics (Newnham et al., 2015). Various studies have used LiDAR for forestry surveying (Huang et al., 2022; Kong et al., 2021), such as the assessment of above-ground biomass (Gonzalez de Tanago et al., 2018), estimation of forest carbon stocks (Saatchi et al., 2011), exploration of vegetation cooling effects (Kong et al., 2022; Kong et al., 2016), and investigation of forest phenology (Nunes et al., 2022). Although LiDAR point clouds provide detailed information on plant structure, they have received little attention in the study of the acoustic effects of vegetation. Existing research primarily concentrates on tree belts rather than single trees (STs) that form the belts. Understanding how STs impact sound propagation based on their 3D characteristics will aid in selecting and arranging appropriate plants for effective noise abatement. Therefore, more detailed and precise structural measurements are needed.

With the support of 3D vegetation information obtained from a handheld scanning LiDAR point clouds and in-situ measured sound pressure level (SPL), this study aims to examine how tree structure impacts sound attenuation. We then established a model considering the fine-scale structural characteristics of trees to enhance the accuracy of existing noise prediction models. The results of this study will be helpful to manage and select trees to reduce noise pollution effectively (Fig. 1). The study was conducted in three phases. First, we investigated the acoustic effects of STs in A-weighted decibels (dBA) and SPL at different frequencies. Second, we extracted morphological parameters of STs from point cloud data to assess the impacts of tree characteristics on sound attenuation. Lastly, we used support vector regression (SVR) to predict the sound attenuation using 3D tree characteristics.

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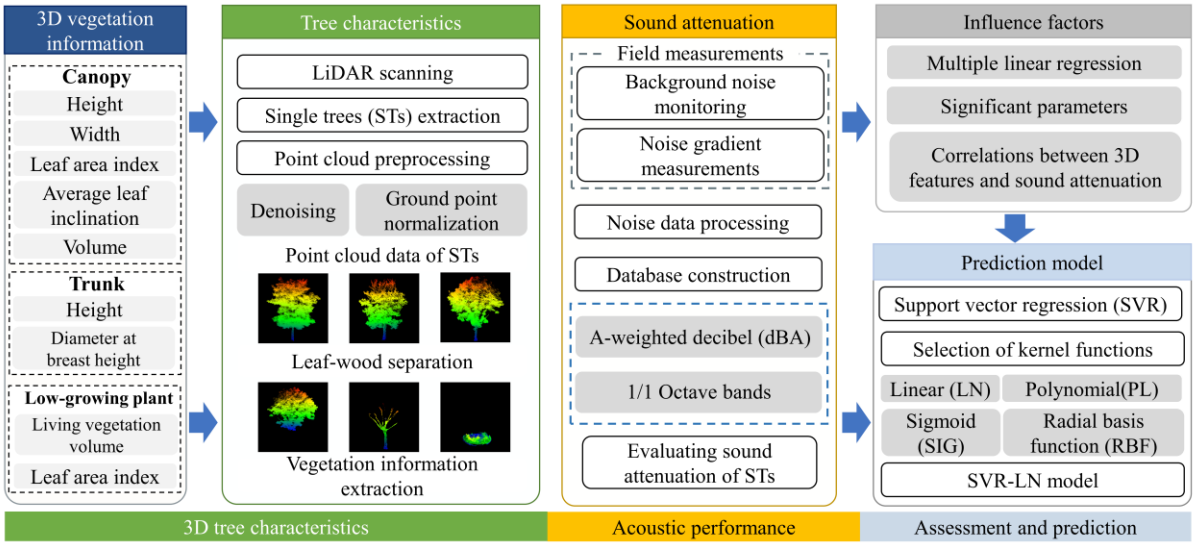


Fig. 1. Flowchart for assessing and predicting the sound attenuation of single trees using accurate 3D tree characteristics derived from a handheld scanning LiDAR.

2 Methods

2.1 Study site and tree selection

This study was conducted at the Xianlin Campus of Nanjing University (Lat: 32°6′53″N, Lon: 118°57′24″E) in Nanjing, Jiangsu Province, China. The Xianlin Campus is adjacent to Yuanhua Road in the east, Metro Line 2 and Xianlin Avenue in the south, Nanjing Bypass Highway and Jiuxianghe East Road in the west, and Weidi Road in the north (Fig. 2a). Enclosed by transportation nodes, the acoustic environment on campus is a complex mix of traffic and other anthropogenic noise. To find the time of day with minimal background noise for our controlled experiment, we monitored the ambient noise in an area with a high concentration of people for 24 hours on September 21 and 22, 2022 (the red star in Fig. 2a). The average daytime noise level (L_{day}) reached 57.1 dBA (Fig. 2b), exceeding the 55 dBA threshold permitted by the Ministry of Ecology and Environment of the People's Republic of China (2008). The nighttime noise level (L_{night}) was 45.9 dBA with an average SPL of 51.7 dBA from 22:00h to 00:00h and 44.0 dBA from 00:00h to 06:00h (Fig. 2c). Based on these measurements, we chose the 00:00h to 06:00h period to conduct the field experiment.

Noise measurements were conducted near 26 trees distributed across six plots on campus. All sample trees were street trees, planted in the soil near the roads and away from buildings, ensuring as consistent experimental conditions as possible. To eliminate the impacts of neighboring trees on the experimental results, we sampled trees with canopy edges away from each other. Sampled species include Deodar Cedar (*Cedrus deodara*), Chinese Hackberry (*Celtis sinensis*), cherry blossoms (*Prunus* subg. *Cerasus*), camphor tree (*Camphora officinarum*), Chinese horse chestnut (*Aesculus chinensis*), Japanese Zelkova (*Zelkova serrata*), and fragrant olive (*Osmanthus fragrans*) (the red circles in Fig. 2a and Appendix A).

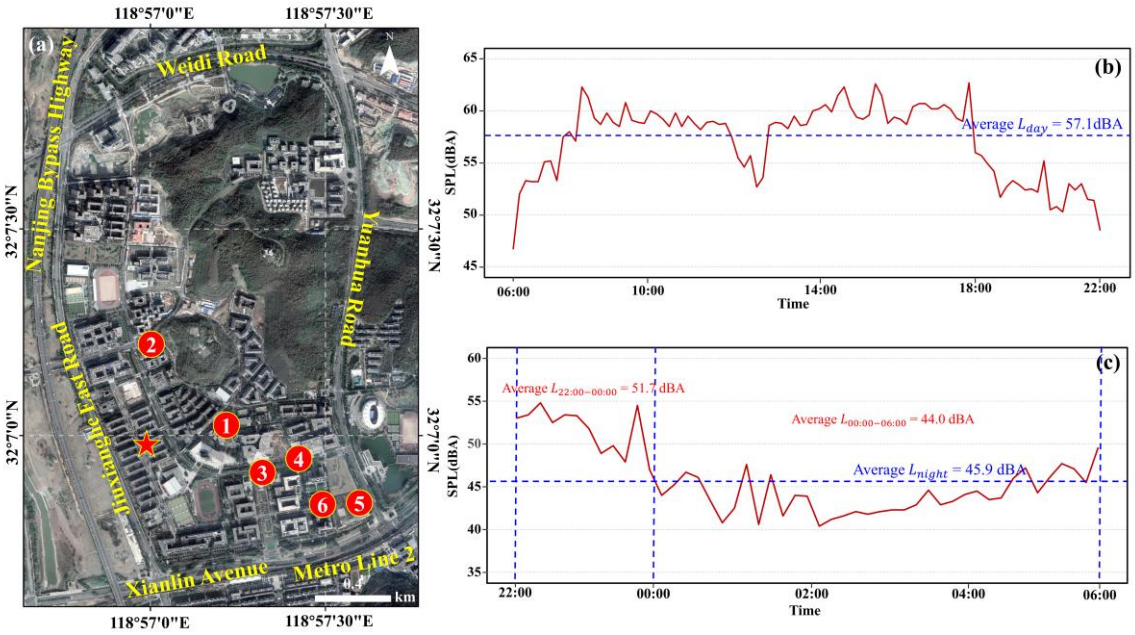


Fig. 2. Study area and background noise. (a) The study area is surrounded by five roads and a subway; the star represents the sampling point for background noise, while the circles represent the field measurement locations. (b) L_{day} was measured from 06:00h to 22:00h on September 21, 2022. (c) L_{night} was measured from 22:00h on September 21, 2022, to 06:00h on September 22, 2022.

2.2 Experimental design and field measurements

2.2.1 Noise field measurements

Given the varying sizes and shapes of the sample trees, we developed a flexible plan for noise measurements. The sound source was placed 1.5 m above the ground at a distance equivalent to the tree height (point S_n in Fig. 3a). The sound level meters were positioned on the source-tree line in front of and behind the sample tree at the height of the crown centroid to measure SPL in dBA and SPL in 1/1 octave bands (point R_n^1 and R_n^2 in Fig. 3a).

White noise was used to specifically analyze sound attenuation at different frequencies attributed to a single tree. A wireless loudspeaker (Marshall Kilburn II, Zound Industries Services Shenzhen Ltd.) was adjusted to produce constant white noise at each sound source point S_n. The sound receiver points R_n^1 were positioned in front of the sample tree to record SPLs of 55 dBA (Environmental Quality Standard for Noise in China (2008) and Environmental noise in Europe (2020)), 60dBA, and 68 dBA (Annual Report on China Environmental Noise Prevention and Control (2008)). White noise was generated with the Adobe Audition CC2018 software (Adobe Systems Incorporated). At the sound receiver points R_n^1 and R_n^2, the SPL was measured with a sound level meter (AWA6292-1, Hangzhou Aihua Instruments Co., Ltd). Successive measurements took 15 s at each point. Noise measurements were conducted at three different directions around each tree (Fig. 3b). Accordingly, nine noise field measurements were conducted for each tree.

To prevent external sound interference, we conducted the field measurements between 00:00h and 06:00h am calm days (Appendix B). A total of 468 training datasets were collected on six days from September 30 to October 22, 2022 (i.e., Sep.30, Oct.1, Oct.2, Oct.12, Oct.20, and Oct.22, 2022), and 60 testing datasets were collected on two days (i.e., Jun.25 and 27, 2023). During our observations, temperatures and relative humidity remained stable, except for lower temperatures on Oct. 12, Oct.20, and Oct.22, 2022.

2.2.2 Calculation for sound attenuation

Atmospheric absorption and sound reflection were not discussed in this study

Atmospheric absorption of sound depends on the frequency, temperature, humidity, and air pressure (ISO9613-1, 1993). During our observations, temperatures and relative humidity remained stable (Appendix B). Moreover, the effect of ground reflection was offset by sound attenuation determined by the difference between SPLs in front of and behind the tree. The sound reflection from buildings has also been neglected due to the significant distance between trees and buildings. Consequently, we did not consider the atmospheric absorption and the sound reflection under similar atmospheric conditions and planting environments (Attenborough, 2007; Yang et al., 2013b).

The total sound attenuation (ΔSPL) by a single tree can be divided into two aspects: insertion loss (IL) and sound attenuation resulting from geometric spreading (ΔL_D) (Equation 1). Through observations, the ΔSPL can be obtained by the SPL difference in front of (SPL_n^1) and behind (SPL_n^2) the sample tree (Equation 2). Therefore, the IL generated by the tree itself, compared with the open ground (free field), can be estimated by the difference between ΔSPL and ΔL_D (Equation 3). ΔL_D follows the inverse square law (Equation 4) (Lee et al., 2008).

$$\Delta SPL = IL + \Delta L_D \quad (1)$$

$$\Delta SPL = SPL_n^1 - SPL_n^2 \quad (2)$$

$$IL = \Delta SPL - \Delta L_D \quad (3)$$

$$\Delta L_D = 20\lg\left(\frac{D_n^2}{D_n^1}\right) \quad (4)$$

where D_n^1 and D_n^2 refers to the distances from the sound source to the front and back receiver points (Fig.3b), respectively; *n* represents the *n*th measurement. In this study, the *D* is the canopy width of each single tree, and the average width of the canopy is 5.25 m.

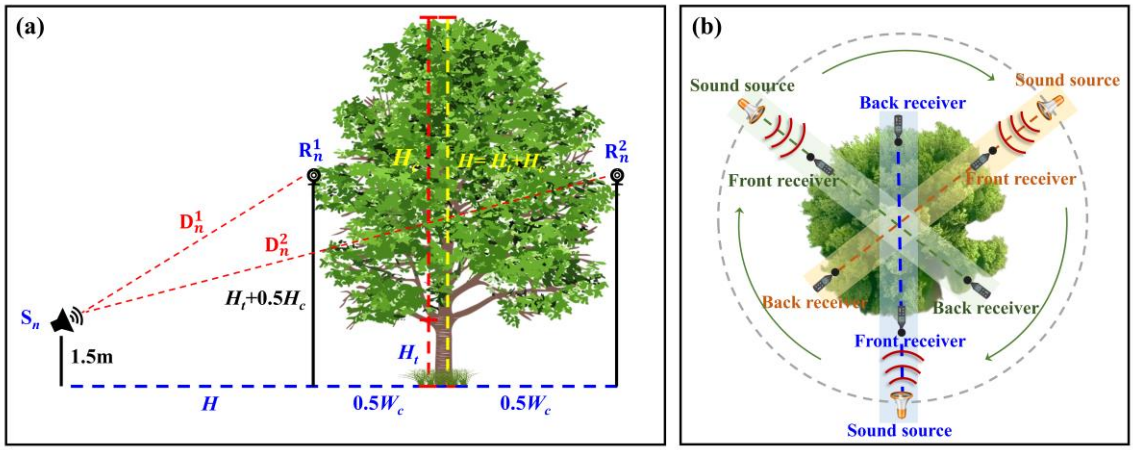


Fig. 3. Experimental design and field measurements: (a) layout plan of sound source point S_n and sound receiver points R_n^1 and R_n^2; (b) noise measurements were conducted at three different directions around each tree. Note that D_n^1 and D_n^2 are the distances between the sound source and sound receiver points R_n^1 and R_n^2, respectively; H refers to the tree height; H_t and H_c are the height of the trunk and canopy, respectively; W_c is the width of the canopy; and n represents the nth measurement.

2.3 Morphological measurements of sample trees

2.3.1 3D laser scanning and point cloud data preprocessing

A handheld scanning LiDAR (LiGrip H120, GreenValley Co. Ltd) obtained the point cloud data at six plots. The TSL device has a 360° horizontal and 280° vertical scanning angle, a detection range of 120 m, and a scanning speed of 320,000 pts/s, enabling the collection of high-accuracy point cloud data (≤3 cm). The scans were conducted on sunny days with calm winds (September 12 and 13, 2022, Appendix B).

Since the point cloud data contained information about all objects near a sampling site, including buildings and other vegetation, the target tree points were extracted manually and then imported into the LiDAR360 software (GreenValley Co. Ltd) for preliminary data preprocessing (i.e., denoising, ground point normalization). Leaf-wood separation was conducted using the LoWoS method by Wang et al. (2020). The processed data were used

to calculate tree characteristic parameters.

2.3.2 Vegetation information extraction

We calculated nine tree characteristics for the 26 sample trees (Table 1). Crown height (H_c), crown width (W_c), diameter at breast height (DBH), and trunk height (H_t) were calculated using LiDAR360; Leaf area index of the crown (LAI_c), average leaf inclination of the crown (ALI), living vegetation volume of low-growing plants below the tree (LVV_L), crown volume (V_c), and LAI of the low-growing plants (LAI_L) were determined using a Python program.

Table 1. Description of tree characteristics.

Vegetation element	Morphological parameters	Description
Canopy	Height (H_c)	Tree crown height is defined as the distance from the lowest branches to the top of the tree.
	Width (W_c)	Canopy width is defined as the distance from one edge to the other, passing through the center of the crown.
	Leaf area index (LAI_c)	LAI is a biophysical parameter representing half the total green leaf area per unit of the horizontal ground surface.
	Average leaf inclination (ALI)	ALI refers to the mathematical description of the angular orientation of plant leaves.
	Volume (V_c)	Canopy volume is a quantitative parameter used to document the 3D size of a tree canopy.
Trunk	Diameter at breast height (DBH)	DBH is the diameter of a tree trunk measured at breast height.
	Height (H_t)	Trunk height is the vertical distance from the base of a tree to the point at which its stem meets the first branch.
Low-growing plant	Living vegetation volume (LVV_L)	LVV_L refers to the total amount of space occupied by low-growing plants.
	Leaf area index (LAI_L)	Same as the LAI mentioned above.

2.3.2.1 Leaf area index

The LAI of the sample trees was identified using the voxel-based canopy profiling (VCP) approach developed by Hosoi and Omasa (2006). The pre-processed point cloud was divided into $i \times j \times k$ grids using Equation (5-7). Grids containing at least one point were labeled 1; the remaining grids were labeled 0. The LAI was then determined by calculating the ratio of the grids labeled 1 to the total number of grids for each plane and

then summarizing the results for all planes, as depicted by Equation (8).

$$i = \mathrm{Int}\left(\frac{X - X_{min}}{\Delta i}\right) \quad (5)$$

$$j = \mathrm{Int}\left(\frac{Y - Y_{min}}{\Delta j}\right) \quad (6)$$

$$k = \mathrm{Int}\left(\frac{Z - Z_{min}}{\Delta k}\right) \quad (7)$$

$$LAI = 1.1 \times \sum_{z=1}^{z=k} \frac{n_1(z)}{n_0(z) + n_1(z)} \quad (8)$$

Where (*i, j, k*) refers to the voxel coordinates within the grids, and (*X, Y, Z*) represents the point cloud coordinates of the LiDAR points. The minimum point cloud values in the x, y, and z directions are denoted by X_{min}, Y_{min} and Z_{min}, respectively, while (Δi, Δj, Δk) signifies the voxel size. $n_1(z)$ and $n_0(z)$ are the grids labeled 1 and 0 at layer k, respectively.

2.3.2.2 Leaf inclination

The ball-pivoting algorithm (BPA) was utilized to generate a triangle mesh that approximates the curved surface of the leaf blade (Bernardini et al., 1999). BPA operates by keeping a ball of radius ρ in contact with two neighboring points and pivoting the ball until it touches the third point to form a triangle (i.e., three points (σ_i, σ_j, σ_k)), in which the ball contains no additional points except those three points. Next, we computed the normal vector *n* (*x, y, z*) of each triangle in the mesh using the function "*triangle_normals*" derived from Open3D for calculating the angle between the normal vector and the z-axis (Zhou et al., 2018), i.e., leaf inclination. The angle corresponding to the leaf inclination between normal vector *n* (*x, y, z*) and z-axis (0, 0, 1) is determined using the following equation:

$$\theta = \arccos \frac{\vec{a} \times \vec{b}}{|\vec{a}| \times |\vec{b}|} \quad (9)$$

where $\vec{a}$ refers to the normal vector *n* (*x, y, z*) and $\vec{b}$ represents the vector (0, 0, 1) corresponding to the z-axis.

2.3.2.3 Vegetation volume

We employed a novel approach named the Layered Area Integral Method (LAIM) to estimate the vegetation volume of the point cloud data (Kong et al., 2016). This method slices the point cloud into equidistant horizontal layers along the z-axis. Each data layer is then projected onto the *x*-*y* plane and transformed into a binary image (blank cells are assigned 0, and others are assigned 1). We then calculate the point cloud area for each layer by summarizing non-empty cells. Finally, the areas of all layers are integrated to obtain the vegetation volume (Equation 10):

$$V = \sum_{1}^{n} (S_i \times h) \quad (10)$$

where h is the height of each layer and S_i is the area of each horizontal point cloud layer.

2.4 Multiple regression

Multiple regression is a statistical technique to analyze the relationship between a single dependent variable and several independent variables. We used multiple regression analysis to investigate how various 3D characteristics of STs impact their acoustic performance. The total sound attenuation for each tree is the dependent variable, and the independent variables are the 3D tree characteristics derived from point cloud data, i.e., H_c, W_c, LAI_c, ALI, V_c, DBH, H_t, LVV_L, and LAI_L.

2.5 Sound attenuation prediction based on support vector regression

2.5.1 Principles of support vector regression

Support vector machine (SVM), a machine learning algorithm rooted in statistical learning theory, has gained widespread popularity across various fields, including pattern recognition, regression analysis, and time series prediction (Kim, 2003; Noble, 2006). SVM can effectively handle high-dimensional data and provide reliable classifications. We utilized Support vector regression (SVR), a highly versatile algorithm based on the concept of SVM, to forecast the sound attenuation potential of STs. SVR aims to find a function

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f(x) that does not deviate more than ε from a target value y_i for all the training data while remaining as flat as possible, as illustrated below (Smola and Schölkopf, 2004):

$$f(x) =< w \cdot x > +b \tag{11}$$

where w denotes the weighted vector, x is the input data point, and b is the bias. Flatness in Equation (11) is a result of a small w. One optimization method is as follows:

$$\begin{aligned} &\text{minimize } \frac{1}{2}\|w\|^2 \\ &\text{subject to } |y_i - f(x_i)| \le \varepsilon \end{aligned} \tag{12}$$

The slack variables ξ_i and ξ_i^* are introduced to allow y_i to deviate within a given range. Equation (12) can be rewritten as follows:

$$\begin{aligned} &\text{minimize } \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*) \\ &\text{subject to } \begin{cases} y_i - f(x_i) \le \varepsilon + \xi_i, i = 1,2\ldots,n \\ f(x_i) - y_i \le \varepsilon + \xi_i^*, i = 1,2\ldots,n \\ \xi_i, \xi_i^* \ge 0, i = 1,2\ldots,n \end{cases} \end{aligned} \tag{13}$$

The constant $C > 0$ determines the trade-off between the flatness of the function f(x) and the amount up to which deviations greater than ε are tolerated. SVR can finally be expressed using the Lagrange Multiplier Method as follows:

$$f(x) = \sum_{i=1}^{n}(\alpha_i - \alpha_i^*)\,K(x_i, x) + b \tag{14}$$

Where α_i and α_i^* are Lagrange multipliers, and $K(x_i, x)$ refers to the kernel function.

2.5.2 Selection of kernel function

The type of kernel function directly impacts the prediction accuracy and is a critical step in the SVR algorithm. Commonly used kernels include linear (LN), sigmoid (SIG), polynomial (PL), and radial basis functions (RBF) (Tehrany et al., 2015). Accordingly, we evaluated four kernel functions to find the one with the highest prediction accuracy: SVR-LN, SVR-SIG, SVR-PL, and SVR-RBF.

2.5.3 Training and testing datasets

SVR requires training and testing data to implement and evaluate a predictive model. To illustrate the universality of the developed model, we constructed the training dataset using the 3D characteristics and sound attenuation of the 26 sample trees. We collected additional testing data on five common street trees: southern magnolia (*Magnolia*

grandiflora), Japanese maple (*Acer palmatum*), fragrant olive (*Osmanthus fragrans*), cherry blossoms (*Prunus* subg. *Cerasus*), and camphor tree (*Camphora officinarum*) (Appendix C).

The point cloud and noise data for the testing trees were collected on June 25th and 27th, 2023 (Appendix B). Since the sound of crickets during the summer might interfere with the accuracy of the results, we increased the source sound levels to 60 dBA and 68 dBA. All subsequent data processing followed the methods described in Section 2.3. Data on tree characteristics was standardized for training and testing, which eliminates the effects of differences in scales and units.

3 Results

First, this section provides an overview of the sample tree characteristics extracted from the point cloud data. Next, we summarize descriptive statistics on the sound attenuation performance of all sample trees by tree species, frequency, and source sound levels. Third, multiple regression results are presented to investigate the main drivers of sound attenuation among 3D tree characteristics. Lastly, we show support vector regression results on predicting sound attenuation induced by STs using their 3D characteristics.

3.1 Sample tree characteristics

The morphological structure of the 26 sample trees varied widely (Table 2), with canopies ranging in height from 3.02 (Tree #17, fragrant olive (*Osmanthus fragrans*)) to 9.67 m (Tree #5, Chinese Hackberry (*Celtis sinensis*)) and width from 3.37 (Tree #18, fragrant olive (*Osmanthus fragrans*)) to 7.29 m (Tree #8, Chinese horse chestnut (*Aesculus chinensis*). Trees also exhibited varying leaf morphologies, with LAI_c ranging from 0.47 (Tree #16, camphor tree (*Camphora officinarum*)) to 4.57 (Tree #12, fragrant olive (*Osmanthus fragrans*)) and *ALI* around 90° (+ 8.37°). The largest tree had a canopy volume of 64.67 m^3 (Tree #7, camphor tree (*Camphora officinarum*)), and the smallest was 2.82 m^3 (Tree #18, fragrant olive (*Osmanthus fragrans*)). Meanwhile, the *DBH* ranged from 0.11 to 0.39 m, and the maximum trunk height was 3.09 m (Tree #16, camphor tree (*Camphora*

officinarum)). The varying leaf density affected the transmissivity of the trees, resulting in vegetation variations underneath, with LVV_L ranging from 0 (Tree #19 and #20, Deodar Cedar (*Cedrus deodara*)) to 2.83 m^3 (Tree #7, camphor tree (*Camphora officinarum*)), and LAI_L ranging from 0 (Tree #19 and #20, Deodar Cedar (*Cedrus deodara*)) to 2.58 (Tree #10, Chinese horse chestnut (*Aesculus chinensis*)).

Table 2. Statistical information regarding tree characteristics of 26 sample trees.

Number (#)	Tree species	Canopy					Trunk		Low-growing plant	
		H_c (m)	W_c (m)	LAI_c	ALI (°)	V_c (m^3)	DBH (m)	H_t (m)	LVV_L (m^3)	LAI_L
1	Deodar Cedar (*Cedrus deodara*)	7.02	5.30	1.21	93.65	14.59	0.20	0.83	0.78	1.63
2	Deodar Cedar (*Cedrus deodara*)	5.81	4.84	2.14	98.37	20.01	0.20	0.87	0.69	1.96
3	Chinese Hackberry (*Celtis sinensis*)	6.71	6.73	2.56	93.89	10.86	0.22	1.65	0.64	2.08
4	cherry blossoms (*Prunus* subg. *Cerasus*)	4.59	5.53	1.57	92.88	15.92	0.26	1.29	1.60	2.39
5	Chinese Hackberry (*Celtis sinensis*)	9.67	6.53	1.56	93.32	37.05	0.22	2.52	0.65	2.26
6	camphor tree (*Camphora officinarum*)	6.40	6.89	1.66	93.97	22.17	0.29	1.66	1.44	2.29
7	camphor tree (*Camphora officinarum*)	7.58	5.38	2.44	93.16	64.67	0.27	2.42	2.83	1.60
8	Chinese horse chestnut (*Aesculus chinensis*)	7.10	7.29	1.23	94.16	25.37	0.25	2.12	1.00	1.85
9	Japanese Zelkova (*Zelkova serrata*)	7.42	6.23	0.76	97.00	44.25	0.29	1.70	0.27	0.89
10	Chinese horse chestnut (*Aesculus chinensis*)	7.02	4.53	2.44	93.89	50.90	0.18	1.19	0.28	2.58
11	fragrant olive (*Osmanthus fragrans*)	4.14	5.10	3.48	92.35	10.94	0.21	1.82	0.49	1.52
12	fragrant olive (*Osmanthus fragrans*)	5.76	5.96	4.57	92.20	17.54	0.23	1.56	0.42	1.32
13	camphor tree (*Camphora officinarum*)	4.84	5.01	1.03	91.18	15.91	0.21	2.46	0.11	2.04
14	camphor tree (*Camphora officinarum*)	3.57	5.01	1.02	92.02	9.59	0.21	2.70	0.12	2.05
15	camphor tree (*Camphora officinarum*)	6.49	6.54	0.84	91.09	59.93	0.23	2.75	0.07	1.15

16	camphor tree (*Camphora officinarum*)	6.85	7.09	0.47	90.97	30.81	0.26	3.09	0.00	0.87
17	fragrant olive (*Osmanthus fragrans*)	3.02	3.46	1.03	91.35	7.55	0.11	0.92	0.12	1.02
18	fragrant olive (*Osmanthus fragrans*)	3.51	3.37	0.69	91.30	2.82	0.11	0.80	0.86	0.94
19	Deodar Cedar (*Cedrus deodara*)	5.14	3.99	0.56	91.32	8.50	0.33	0	0	0
20	Deodar Cedar (*Cedrus deodara*)	5.23	3.87	1.29	91.38	20.02	0.39	0	0	0
21	fragrant olive (*Osmanthus fragrans*)	4.00	4.48	0.83	91.86	16.26	0.23	1.24	0.15	0.88
22	fragrant olive (*Osmanthus fragrans*)	3.20	4.49	0.72	91.78	15.12	0.27	1.49	0.11	0.85
23	fragrant olive (*Osmanthus fragrans*)	5.13	4.89	1.12	91.67	14.48	0.31	1.31	0.11	1.02
24	fragrant olive (*Osmanthus fragrans*)	3.22	4.04	0.86	91.59	12.21	0.24	1.18	0.21	0.99
25	fragrant olive (*Osmanthus fragrans*)	3.94	4.63	0.82	91.51	17.60	0.34	0.82	0.13	0.73
26	fragrant olive (*Osmanthus fragrans*)	4.53	5.30	1.30	91.33	23.94	0.31	0.97	0.20	0.80

3.2 Sound attenuation of single trees

The average IL generated by a single tree was approximately 2-3 dBA, while the total sound attenuation ranged from 6 to 8 dBA when integrating the contribution of geometric spreading within the canopy. We calculated the average SPL at each frequency for three source sound gradients to investigate the sound attenuation at different frequencies in front of and behind trees. The variations of SPL in 1/1 octave bands revealed that sound attenuation was positively correlated with frequency (Fig. 4). The sampled trees reduced sound above 31.5 Hz, with sound attenuation ranging from 3.1 to 12.7 dB, 4.3 to 12.8 dB, and 5.0 to 12.6 dB for frequencies ranging from 500 to 16000 Hz, respectively. The sound attenuation effectiveness varied among the 26 tree samples, but the trees yielded a similar sound attenuation across frequencies (Fig. 5).

Table 3. Sound attenuation of 26 sample trees.

Number (#)	Tree species	ΔSPL (dBA)				IL (dBA)			
		55 dBA	60 dBA	68 dBA	Average	55 dBA	60 dBA	68 dBA	Average
1	Deodar Cedar (*Cedrus deodara*)	5.1	5.5	5.8	5.5	1.0	1.4	1.7	1.3
2	Deodar Cedar (*Cedrus deodara*)	7.3	8.2	8.3	7.9	2.9	3.8	3.9	3.5
3	Chinese Hackberry (*Celtis sinensis*)	5.1	6.3	6.1	5.8	0.4	1.6	1.4	1.2
4	cherry blossoms (*Prunus* subg. *Cerasus*)	5.8	6.4	6.8	6.3	0.4	1.0	1.4	0.9
5	Chinese Hackberry (*Celtis sinensis*)	5.8	5.8	6.6	6.1	2.6	2.6	3.4	2.8
6	camphor tree (*Camphora officinarum*)	6.3	6.3	6.6	6.4	1.4	1.4	1.7	1.5
7	camphor tree (*Camphora officinarum*)	5.6	6.2	6.7	6.2	2.3	2.9	3.4	2.9
8	Chinese horse chestnut (*Aesculus chinensis*)	6.7	8.3	8.5	1.8	2.2	3.8	4.0	3.3
9	Japanese Zelkova (*Zelkova serrata*)	5.8	6.2	6.5	6.2	1.7	2.1	2.4	2.1
10	Chinese horse chestnut (*Aesculus chinensis*)	5.3	5.8	6.0	5.7	1.8	2.3	2.5	2.2
11	fragrant olive (*Osmanthus fragrans*)	7.4	9.0	9.1	8.5	2.5	4.1	4.2	3.6
12	fragrant olive (*Osmanthus fragrans*)	8.4	10.2	10.8	9.8	3.7	5.5	6.1	5.1
13	camphor tree (*Camphora officinarum*)	7.8	8.0	8.5	8.1	3.8	4.0	4.5	4.1
14	camphor tree (*Camphora officinarum*)	6.1	6.6	7.4	6.7	1.6	2.1	2.9	2.2
15	camphor tree (*Camphora officinarum*)	6.3	7.1	7.4	6.9	2.2	3.0	3.3	2.9
16	camphor tree (*Camphora officinarum*)	6.8	7.0	7.3	7.0	2.7	2.9	3.2	3.0
17	fragrant olive (*Osmanthus fragrans*)	10.6	11.7	12.0	11.4	5.3	6.4	6.7	6.1
18	fragrant olive (*Osmanthus fragrans*)	7.0	7.6	7.7	7.4	2.2	2.8	2.9	2.6
19	Deodar Cedar (*Cedrus deodara*)	7.0	7.6	8.4	7.7	2.1	2.7	3.5	2.8
20	Deodar Cedar (*Cedrus deodara*)	7.4	7.5	8.1	7.7	2.7	2.8	3.4	3.0
21	fragrant olive (*Osmanthus fragrans*)	7.0	7.5	7.4	7.3	2.0	2.5	2.4	2.3
22	fragrant olive (*Osmanthus fragrans*)	7.9	9.0	8.9	8.6	2.4	3.5	3.4	3.1
23	fragrant olive (*Osmanthus fragrans*)	9.1	9.6	9.8	9.5	4.6	5.1	5.3	5.0
24	fragrant olive (*Osmanthus fragrans*)	7.1	7.6	7.6	7.4	1.7	2.2	2.2	2.0
25	fragrant olive (*Osmanthus fragrans*)	7.6	8.6	8.7	8.3	1.9	2.9	3.0	2.6
26	fragrant olive (*Osmanthus fragrans*)	8.1	8.8	9.2	8.7	2.5	3.2	3.6	3.1

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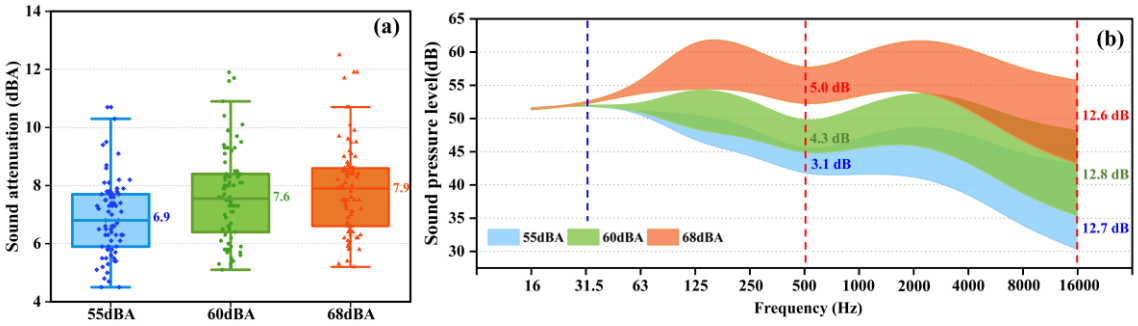


Fig. 4. Overall sound attenuation for 26 sample trees: (a) average sound attenuation corresponding to three source sound levels, and (b) sound attenuation at various frequencies.

The sound gradient measurements averaged by tree species showed sound attenuation differently (Deodar Cedar (*Cedrus deodara*), 7.2 ± 0.5 dBA, Chinese Hackberry (*Celtis sinensis*), 6.1 ± 0.6 dBA, camphor tree (*Camphora officinarum*), 6.9 ± 0.4 dBA, Chinese horse chestnut (*Aesculus chinensis*), 7.1 ± 1.0 dBA, fragrant olive (*Osmanthus fragrans*), 9.0 ± 1.0 dBA, and others (including cherry blossoms (*Prunus* subg. *Cerasus*) and Japanese Zelkova (*Zelkova serrata*)), 6.3 ± 0.5 dBA) (Fig. 5a). The SPL in the 1/1 octave bands for all tree species showed a consistent pattern, increasing with frequency and following the overall trend. Sound attenuation levels peaked at around 125 Hz, then slightly decreased until 500 Hz before resuming their upward trend at higher frequencies (Fig. 5b-5g).

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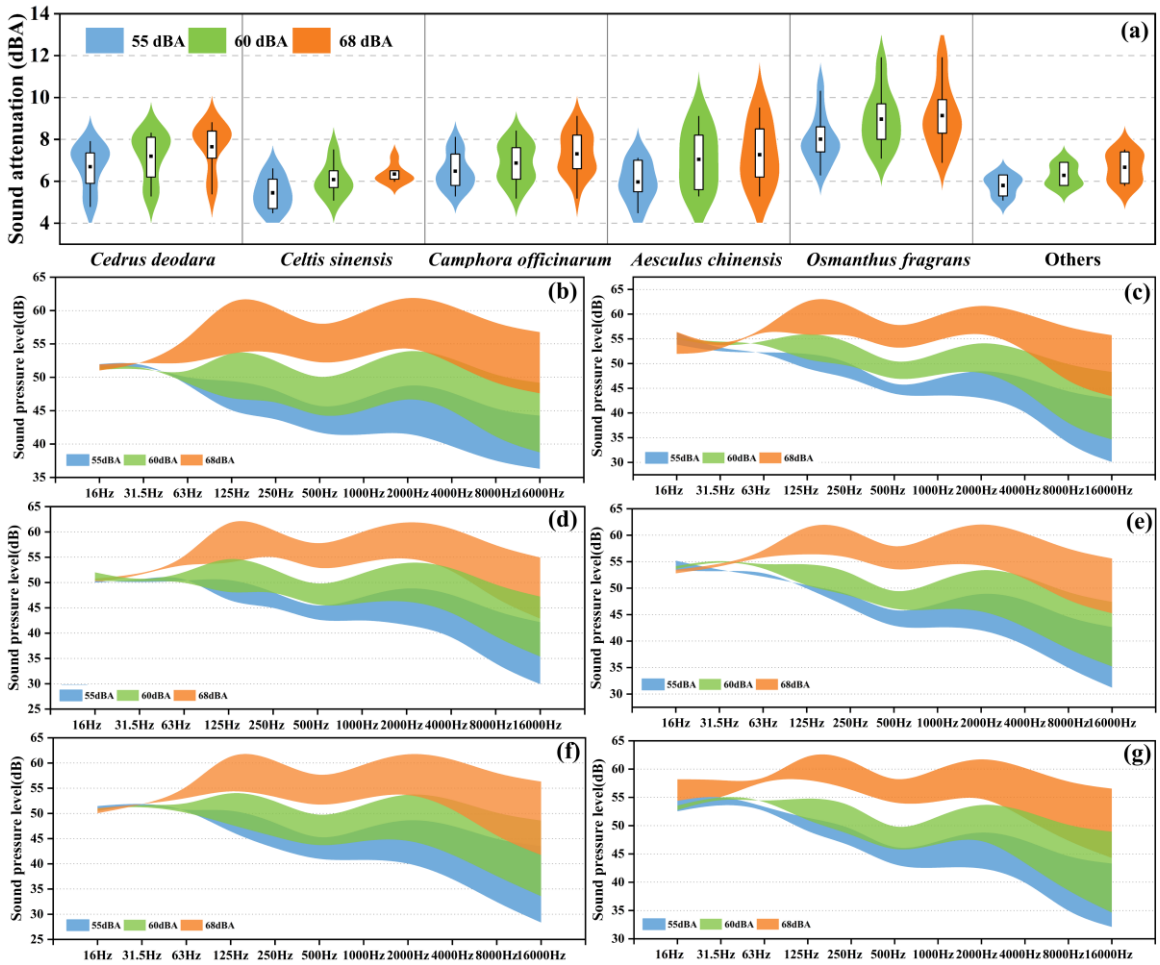


Fig. 5. Sound attenuation by tree species: (a) sound attenuation for all tree species, (b) Deodar Cedar (*Cedrus deodara*), (c) Chinese Hackberry (*Celtis sinensis*), (d) camphor tree (*Camphora officinarum*), (e) Chinese horse chestnut (*Aesculus chinensis*), (f) fragrant olive (*Osmanthus fragrans*), and (g) others (including cherry blossoms (*Prunus* subg. *Cerasus*) and Japanese Zelkova (*Zelkova serrata*)).

STs provide consistent effectiveness in sound attenuation, determined by their 3D characteristics rather than sound exposure levels. We calculated the sound attenuation proportional for each source sound level (55dBA, 60dBA, and 68dBA) by dividing the respective attenuation values by the sum of all attenuation values obtained from the three sound gradient measurements. Our findings show that the reduction proportion for each

tree was approximately 33% (Tree #12, fragrant olive (*Osmanthus fragrans*) has a maximum of 37%, and Tree #8, Chinese horse chestnut (*Aesculus chinensis*), has a minimum of 28%) (Fig. 6). Combining the results from Fig. 4 to Fig. 6, there seems to be greater sound attenuation by STs at higher exposure levels. This phenomenon could be attributed to foliage absorbing sound by converting sound energy into heat energy through the mechanical vibration of leaves (Li and Kang, 2020). Exposed to higher SPLs corresponding to increased sound pressures, there are more mechanical vibrations of foliage and enhanced sound absorption, resulting in greater sound attenuation.

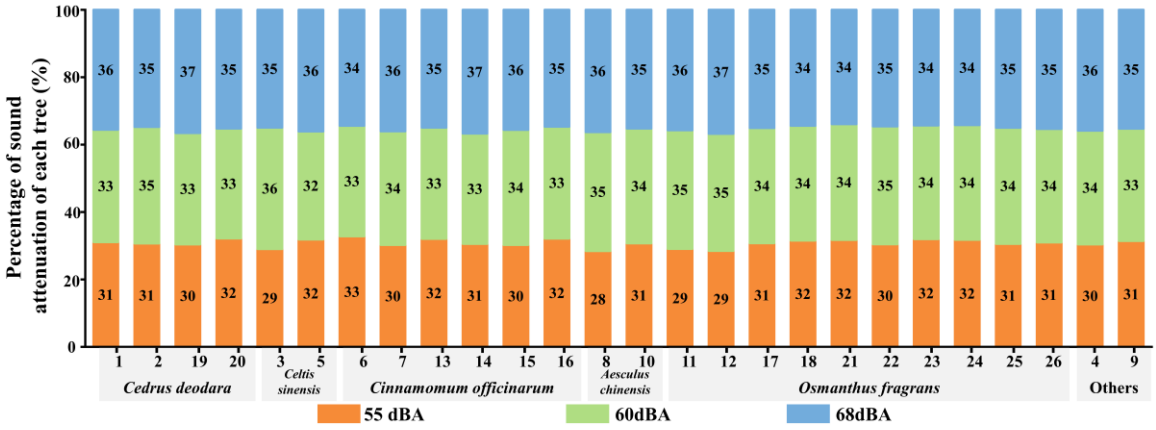


Fig. 6. The sound attenuation for each tree corresponding to the three source sound levels.

3.3 Correlations between tree characteristics and sound attenuation

Using multiple linear regression analysis, we investigated the statistical relationship between sample tree characteristics (independent variables) and the total sound attenuation (dependent variable). All variables in the study passed the hypothesis test, with a Durbin Watson (DW) value of 1.753 and variance inflation factors (VIF) below 5. The regression analysis revealed that H_c, LAI_c, ALI, and LAI_L are the main drivers of sound attenuation (Table 4). LAI_c has a strongly positive coefficient ($\beta = 0.332$, $p < 0.01$), suggesting that it is the most important sound attenuation factor, with ALI of the canopy also having an advantageous effect ($\beta = 0.168$, $p < 0.01$). LAI_L and H_c, conversely, have negative

correlations with sound attenuation (β = -0.438, p < 0.01, and β = -0.295, p < 0.01, respectively). Other tree characteristics are not statistically significant, implying these morphological characteristics do not strongly influence sound attenuation.

Table 4. Multiple linear regression results for sound attenuation as dependent variable and tree characteristics as independent variables.

Variable	Unstandardized Coefficients B	Standardized Coefficients β	(n=234, R^2=0.663, DW=1.753) t	Sig.	VIF
Constant	1.093		0.354	0.723	
H_c**	-0.278	-0.295	-2.849	0.005	4.297
W_c	0.040	0.028	0.255	0.799	4.788
LAI_c**	0.547	0.332	5.841	0.000	1.288
*ALI***	0.100	0.168	3.046	0.003	1.222
V_c	-0.014	-0.163	-1.874	0.062	3.041
DBH	-3.167	-0.127	-1.850	0.066	1.888
H_t	0.282	0.143	1.518	0.130	3.533
LVV_L	0.249	-0.100	-1.623	0.106	1.530
LAI_L**	-0.989	-0.438	-5.578	0.000	2.460

*Indicates statistical significance at the 5% level (two-tailed).
*Indicates statistical significance at the 1% level (two-tailed).

3.4 Prediction for sound attenuation using relevant tree characteristics

SVR was used to predict the total sound attenuation considering statistically significant tree characteristics, i.e., H_c, LAI_c, *ALI*, and LAI_L. The four kernels SVR-LN, SVR-SIG, SVR-PL, and SVR-RBF yielded a confidence of 77.2%, 44.7%, 15.4%, and 58.3%, respectively (Table 5). SVR-LN had the highest confidence, indicating it is the most appropriate model for accurately predicting sound attenuation by STs.

Table 5. Confidences of four SVR kernel types.

Kernel	Linear	Sigmoid	Polynomial	RBF
Confidence	77.2%	44.7%	15.4%	58.3%

Comparing predicted and observed values of the testing dataset (Table 6) shows a good model fit (R^2=0.74, RMSE=0.38, and MSE=0.15), which suggests that predicting the

sound attenuation based on 3D tree characteristics is feasible. The predictive SVR-LN model has the potential to predict the sound attenuation of trees in various urban settings, providing valuable information for urban planners and policymakers when making decisions related to the management and maintenance of urban soundscapes.

Table 6. Detailed information regarding sound attenuation and 3D characteristics of five trees (testing data).

Tree species	H_c (m)	LAI_c	ALI (°)	LAI_L	Total sound attenuation (dBA)	
					Actual values	Predicted values
southern magnolia (*Magnolia grandiflora*)	6.43	1.44	89.97	0.32	9.5	9.4
Japanese maple (*Acer palmatum*)	2.30	1.30	95.80	1.52	8.0	8.1
fragrant olive (*Osmanthus fragrans*)	2.84	2.97	93.52	0.17	10.2	10.5
cherry blossoms (*Prunus* subg. *Cerasus*)	3.23	1.64	96.72	0.50	8.6	9.3
camphor tree (*Camphora officinarum*)	5.32	2.49	90.67	0.61	9.3	9.6

4 Discussion

4.1 The contributions of single trees to urban noise reduction

Trees play an important role in urban soundscapes, and single trees can provide an IL of 2-3 dBA and a total sound attenuation of 6-8 dBA, attributed to their 3D characteristics. Similar outcomes have been achieved when considering a single tree as a tree belt with a restricted depth of approximately 5-6 m. For example, a pioneering study by Kragh (1981) on the acoustic effects of tree belts revealed that a 3 m deep tree belt exhibited an IL of 0-1 dBA, while a dense, mature coniferous tree belt achieved an IL of 3-5 dBA. Fang and Ling (2005) reported a noise reduction of approximately 3 dBA for a *Casuarina equisetifolia* tree belt with a depth of 3 m. Furthermore, Ow and Ghosh (2017) found a noise reduction of 2 dBA for a sparsely to moderately planted tree belt compared to open

ground. Our findings are in line with other studies' results that trees are more effective at attenuating sound waves with middle to high frequencies (>500Hz) than low-frequency sound waves (Gaudon et al., 2022; Pal et al., 2000; Tyagi et al., 2006; Yang, 2013). We observed a sound attenuation pattern that starts at 31.5 Hz, peaks locally between 125 and 250 Hz, slightly decreases until 500 Hz, and then displays a progressive increase (Fig.4 and 5). Our experimental results highlight the significant contribution of individual trees to localized noise reduction, e.g., planting single trees in front of apartment windows, particularly for areas consistently exposed to high-frequency noise; however, since a single tree is the smallest element of tree belts and other green spaces, the evaluation of the effectiveness and characteristics of a single tree is essential for comprehensive noise reduction strategies through the implementation of vegetation belts in places where the acoustic environment cannot meet the noise standard, e.g., in residential areas, at schools, hospitals, etc. Therefore, our observations suggest that strategic tree planting could potentially reduce average traffic noise levels from 68 dBA to meet the recommended threshold of 60 dBA set by the "China Environmental Noise Prevention and Control (2008)".

4.2 Increasing sound attenuation of single trees through denser canopy

Tree canopy plays a predominant role in sound attenuation, and the LAI_c is the most crucial parameter that determines the acoustic performance of STs. The above-ground part of a single tree includes a trunk and the canopy comprising branches and abundant foliage. Compared to tree belts composed of numerous trees, contributions from the canopy on sound attenuation are more significant than those from trunks and forest floors. A study by Yang et al. (2013b) who quantified scattered sound energy from a single tree in field experiments using reverberation time also highlighted the importance of the tree crown. The interaction between sound waves and vegetation elements is a complex process that is affected by multiple factors. Our findings suggest that H_c, LAI_c, ALI, and LAI_L are the main determinants in reducing noise. These results are supported by prior observations that

increased LAI_c and *ALI* of trees reduce noise (Horoshenkov et al., 2013; Karbalaei et al., 2015; Xu et al., 2022). The leaf inclination determines the effective path length through which the incident sound waves propagate. Larger leaf inclination means that more leaves are perpendicular to the direction of the incident sound waves, resulting in greater sound absorption by leaves due to thermo-viscous effects and damped vibrations. *LAI*, defined as the one-sided leaf area per unit of horizontal ground surface area (Weiss et al., 2004), quantifies the leaf amount in a canopy. Canopies with higher LAI typically have more leaves and fewer gaps through which incident sound waves can propagate. In contrast, regression results indicated a negative impact of canopy height on the acoustic effects of STs. A higher canopy usually exhibits more gaps through which more sound waves penetrate while the leaf area index remains constant. Lastly, we found that trees with greater leaf inclination are more effective in reducing noise. However, controlling leaf inclination in street trees is challenging, as it depends on plant physiology and is determined by plant hormones regulated by environmental stresses (Xu et al., 2021). Noise reduction policies on street trees should focus on the leaf area density instead, which is easier to control.

4.3 Limitations and prospects

Our work has several limitations concerning the data collection, analysis, and interpretation of results. First, our tree sampling protocol restricted the number of trees eligible for the study. Sample trees had to have canopies apart from neighboring trees and be located away from the residential areas on campus due to nighttime noise measurement requirements. These restrictions resulted in a smaller sample size of the STs. We selected different tree species commonly planted in Nanjing to explore the influence of tree structure characteristics on sound attenuation. However, limited by the number of samples for each species (e.g., *Osmanthus fragrans* has ten samples, and *Prunus* subg. *Cerasus* has one sample), we cannot provide recommendations on the best species for noise reduction. Future studies should assess the typical sound attenuation of various tree species using larger sample sizes and further investigate the enhanced sound attenuation by STs at higher

exposure levels, also considering the sensitivity or tolerance to other urban stressors (e.g., heat and drought). Second, our study did not investigate seasonal effects. Our findings revealed that most of the sound attenuation by STs stems from the canopy, and leaves play a crucial role in acoustic performance due to interactions between the leaves and sound waves. However, deciduous trees exhibit variations in leaf biomass across different seasons, leading to inevitable changes in sound attenuation from the peak foliage to the leaf-off season. Further research is needed to investigate the seasonal acoustic performance of STs under varying leaf biomass conditions to develop more accurate and effective noise regulation strategies. Third, we found a statistically significant inverse relationship between LAI_L and sound attenuation. This result contradicts findings from another study reporting that shrubs enable trees to provide better sound attenuation (Fang and Ling, 2003; Van Renterghem et al., 2012) and requires further data collection and study.

5 Conclusions

The findings of this study show that the 3D morphological characteristics of a single tree influence sound propagation, and a single tree can reduce urban noise. Handheld LiDAR scans and acoustic field measurements were conducted to explore the sound attenuation of STs and the driving morphological and foliage characteristics. The impact of tree characteristics on sound attenuation varied, resulting in IL and total sound attenuation ranging from 2 to 3 dBA and 6 to 8 dBA, respectively. The most significant reduction was observed above 500 Hz. Of interest here was that sound attenuation by STs slightly increased with higher source sound levels. Regression analysis showed that trees with higher LAI_c provided more sound attenuation benefits, which has practical implications as it directly informs the choice of appropriate tree species to alleviate urban noise problems and improve urban soundscapes. Utilizing trees to reduce noise is constrained by the requirement for more extensive spaces compared with rigid noise barriers, which creates a trade-off with available space in rights-of-way. Therefore, it is

recommended to use small trees with high leaf density for effective and comprehensive noise abatement.

The SVR-LN model based on tree morphology and foliage characteristics predicted sound attenuation induced by STs well (R^2=0.74, RMSE=0.38, and MSE=0.15). Our prediction approach facilitates the evaluation of urban soundscapes at pedestrian-relevant scales and can guide environmental noise management in diverse urban contexts. The developed predictive model, if integrated into comprehensive noise mapping strategies, will be helpful for urban planning and policymaking to evaluate, manage, and mitigate urban noise pollution.

CRediT authorship contribution statement

Jian Lu: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. Fanhua Kong: Writing – original draft, Supervision, Software, Resources, Project administration, Methodology, Investigation, Conceptualization. Haiwei Yin: Writing – review & editing, Resources, Methodology, Data curation. Ariane Middel: Writing – review & editing. Jian Kang: Writing – review & editing. Zhihao Wen: Investigation. Hongqing Liu: Investigation.

Data availability

Data will be made available on request.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Appendix A. Supplementary data Supplementary data to this article can be found online at https://doi. org/10.1016/j.jenvman.2024.122818.

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